

Note 2.6 Limits Involving Infinity; Asymptotes of Graphs

Infinity Notation

- $x \rightarrow +\infty$: x increases without bound.
- $x \rightarrow -\infty$: x decreases without bound.
- $x \rightarrow \infty$: $|x|$ increases without bound.

Finite limits as $x \rightarrow \pm\infty$

- **Definition:**

- We say $f(x)$ has the limit L as $x \rightarrow +\infty$ and write

$$\lim_{x \rightarrow +\infty} f(x) = L$$

if for any number $\epsilon > 0$, there exists a corresponding positive number M such that

$$x > M \implies |f(x) - L| < \epsilon$$

- and

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if for any number $\epsilon > 0$, there exists a corresponding positive number N such that

$$x < N \implies |f(x) - L| < \epsilon$$

- and

$$\lim_{x \rightarrow \infty} f(x) = L$$

if for any number $\epsilon > 0$, there exists a corresponding positive number A such that

$$|x| > A \implies |f(x) - L| < \epsilon$$

- **Theorem:**

$\lim_{x \rightarrow \infty} f(x) = L$ if and only if $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = L$

- Corollary:

- If $\lim_{x \rightarrow +\infty} f(x) \neq \lim_{x \rightarrow -\infty} f(x)$, then $\lim_{x \rightarrow \infty} f(x)$ doesn't exist.
 - $\lim_{x \rightarrow \infty} \frac{1}{x^n} = \lim_{x \rightarrow +\infty} \frac{1}{x^n} (\alpha > 0) = \lim_{x \rightarrow -\infty} \frac{1}{x^\alpha} (\alpha > 0, x^\alpha \text{ is defined at negative values})$

Limits at Infinity of Rational Functions and Radical Functions

- **Rational Functions:**

First divide the numerator and denominator by the highest power of x in the denominator.

- **Radical Functions:**

Multiply the numerator and denominator by conjugate radical expression.

- **Remarks:**

- $$\lim_{x \rightarrow \infty} \frac{a_0x^m + a_1x^{m-1} + \dots + a_m}{b_0x^n + b_1x^{n-1} + \dots + b_n} = \begin{cases} \frac{a_0}{b_0}, n = m \\ 0, n > m \\ \infty, n < m \end{cases}$$
- and

$$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)} = +\infty (\text{or } -\infty, \infty)$$

when $\lim_{x \rightarrow x_0} P(x) \neq 0, \lim_{x \rightarrow x_0} Q(x) = 0$.

- **Horizontal Asymptotes:**

If $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow +\infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$, then $y = b$ is a horizontal asymptote of the graph of $f(x)$.

- **Theorem:**

If $\lim f(x) = 0, g(x)$ is a bounded function, then $\lim f(x)g(x) = 0$

e.g: $\lim_{x \rightarrow +\infty} \frac{1}{x} \sin x = 0$

- **Oblique Asymptotes:**

- Definition:

If $\lim_{x \rightarrow +\infty} (f(x) - ax - b) = 0$ or $\lim_{x \rightarrow +\infty} (f(x) - ax - b) = 0$ or

$\lim_{x \rightarrow \infty} (f(x) - ax - b) = 0 (a \neq 0)$, then the line $y = ax + b$ is the oblique asymptotes of the graph of $f(x)$. (When the degree of numerator greater than denominator in a rational function, it will have an oblique asymptote)

- Theorem:

The graph of $f(x)$ has an oblique asymptote if and only if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = a \neq 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} (f(x) - ax) = b (\text{or } x \rightarrow +\infty, x \rightarrow -\infty)$$

Infinite Limits

- **Definition:**

$$\lim_{x \rightarrow c^+} f(x) = \infty \Leftrightarrow \text{For any } M > 0, \text{ there exists a } \delta > 0, \text{ when } x \in (c, c + \delta), |f(x)| > M$$

$$\lim_{x \rightarrow c^-} f(x) = \infty \Leftrightarrow \text{For any } M > 0, \text{ there exists a } \delta > 0, \text{ when } x \in (c - \delta, c), |f(x)| > M$$

$$\lim_{x \rightarrow c} f(x) = \infty \Leftrightarrow \text{For any } M > 0, \text{ there exists a } \delta > 0, \text{ when } x \in (c - \delta, c + \delta), |f(x)| > M$$

Vertical Asymptotes

- **Definition:**

- If $\lim_{x \rightarrow a^+} f(x) = \infty$ or $\lim_{x \rightarrow a^-} f(x) = \infty$, then $x = a$ is called the vertical asymptotes of $f(x)$. (Sometimes, a is the endpoint of its domain)

The Limit of a Sequence $\{a_n\}$

- **Find limits:**

The sequence $\{a_n\}$ is a special function, which is $f(n) = a_n, n \in N^+$, thus the method of finding limit $\lim_{n \rightarrow \infty} a_n$ is similar to $\lim_{x \rightarrow \infty} f(x)$.

- **Monotone Convergence Theorem:**

- $f(x)$ is increasing when $x < c, f(x) \leq M$ for some M , then $\lim_{x \rightarrow c^-} f(x)$ exists.
- $f(x)$ is decreasing when $x > c, f(x) \geq L$ for some L , then $\lim_{x \rightarrow c^+} f(x)$ exists.

For $\{a_n\}$, if $a_n \leq a_{n+1} (a_n \geq a_{n+1})$ when $n \geq n_0$ and $a_n \leq M (a_n \geq L)$ for all n , then $\lim_{n \rightarrow \infty} a_n$ exists.

- **Important Limits:**

- $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$
- $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$
- $\lim_{x \rightarrow 0} \left(1 + \frac{1}{x}\right)^x = e$

Generally,

- $\lim_{f(x) \rightarrow \infty} \left(1 + \frac{1}{f(x)}\right)^{f(x)} = e$
- $\lim_{f(x) \rightarrow 0} (1 + f(x))^{f(x)} = e$

- **Corollary:**

If $\lim_{x \rightarrow c} u(x) = a (a > 0), \lim_{x \rightarrow c} v(x) = b$, then $\lim_{x \rightarrow c} u(x)^{v(x)} = a^b$